

# MA 442 - Quiz

April 1

**Name:** \_\_\_\_\_ **BUID:** \_\_\_\_\_

**Question 1.** Use elementary row operations to compute the determinant of

$$B = \begin{bmatrix} 1 & 1 & 0 & 2 \\ 2 & 3 & 1 & 5 \\ 1 & 0 & 2 & 1 \\ 3 & 4 & 1 & 8 \end{bmatrix}. \quad (1)$$

(Hint: You can stop performing row operations when you get an upper triangular matrix. Why?)

**Question 2.** Suppose that

$$\varphi: \mathbb{R}^n \times \cdots \times \mathbb{R}^n \rightarrow \mathbb{R} \quad (2)$$

is an  $n$ -linear, alternating function. Show that

$$\varphi(\mathbf{v}_1, \dots, \mathbf{v}_n) = \varphi(\mathbf{v}_1 - \mathbf{v}_n, \mathbf{v}_2 - \mathbf{v}_n, \dots, \mathbf{v}_{n-1} - \mathbf{v}_n, \mathbf{v}_n). \quad (3)$$

for any vectors  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^n$ .

